The empirical recurrence for OEIS A224350: recovering a conjecture that is not written down

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Abstract

OEIS A224350 records "Empirical recurrence of order 56", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 242$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 56, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 56 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A224350 is "Number of nX5 0..2 arrays with diagonals and antidiagonals unimodal and rows nondecreasing.". It has offset 1 and begins

$$21, 441, 5840, 57387, 495985, 4231138, 37433596, 342170839, \ \dots$$

The entry states:

> Empirical recurrence of order 56 (see link above)

As of the "Last modified" line on the live entry (Oct 03 2025 23:23:24, revision 10) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 21504$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 242$ of the 21504, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 242$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 484$ exact terms of the model returns order 56 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{56} c_i\, a(n-i),$$

c_1	21	c_{15}	206116775	c_{29}	−16676275105303	c_{43}	−189711176803738
c_2	−141	c_{16}	−1594661918	c_{30}	60665922201557	c_{44}	180737037916438
c_3	225	c_{17}	603671235	c_{31}	146409336245365	c_{45}	−14469409442964
c_4	1321	c_{18}	−17184166928	c_{32}	152192236606539	c_{46}	24639998343684
c_5	−6843	c_{19}	47512805588	c_{33}	166891220103368	c_{47}	92780756084192
c_6	11044	c_{20}	21967118302	c_{34}	33883719620933	c_{48}	−52473842214432
c_7	−19994	c_{21}	28273266639	c_{35}	−501951028899368	c_{49}	−54680942134096
c_8	74540	c_{22}	631120518544	c_{36}	−698534050402451	c_{50}	−12828955816800
c_9	−47303	c_{23}	405623824038	c_{37}	99987275858170	c_{51}	−9044864396656
c_{10}	2874390	c_{24}	−621215559718	c_{38}	276124082759842	c_{52}	−909464003904
c_{11}	−6954443	c_{25}	−3829360905075	c_{39}	−178714259267186	c_{53}	3762194817088
c_{12}	−22939637	c_{26}	−6520940310132	c_{40}	129707190962539	c_{54}	2336700690432
c_{13}	88753600	c_{27}	−19765485108578	c_{41}	491840845101631	c_{55}	411390210048
c_{14}	−117057398	c_{28}	−20635164386656	c_{42}	−38486545906869	c_{56}	350355456

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 56, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $56 + 4$ terms removed returns order 56 as well, so $\deg q_{\min} = 56 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $56+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 56 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A224350.

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